Practice workbook · Describe how heart disease is prevented and treated

Lifestyle and diet changes, statins, stents, bypass surgery and transplant problems, including judging data from a statin study.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-HEART-DISEASE-PREVENTION-AND-TREATMENT

1. Question 4 · part a · 2 marks

November 2023 Paper 2

Worked-example retry · Attempt part a only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Atorvastatin belongs to a group of medicines known as statins. Statins are medicines that regulate lipid levels in the blood. Statins are used when changes of diet and lifestyle have failed to lower blood cholesterol levels.

Statins can also be used to reduce the risk of heart disease when cholesterol levels are normal. People with liver disease or kidney problems should avoid taking statins. Women who are pregnant or are breastfeeding should also avoid taking statins.

In the United Kingdom approximately 7.5 million people take statins. This is 12% of the total population. However, up to 20% of these people stop taking statins because of adverse side effects such as muscle aches, fatigue, feeling sick and joint pains.

A recent study looked at patients who had stopped taking statins because of side effect symptoms. Patients were asked to take statins for four months, then placebo tablets (tablets that look like statins but contain no active substances) for four months followed by no tablets for four months. Every day the patients were asked how bad their symptoms were using a score from 0 to 100. The table shows the results.

Four-month treatment	Mean score out of 100
statins	16.3
placebo	15.4
no tablets	8.0

(a) Describe the changes in diet and lifestyle that could lower cholesterol. (Lines 2 and 3)

(2)

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1. Question 4 · part a · 2 marks

November 2023 Paper 2

Worked-example retry · Attempt part a only.

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(b) (i) Explain why people with liver disease or kidney problems should avoid taking statins. (Lines 5 and 6)

(4)

(ii) Explain why women who are pregnant or breastfeeding should avoid taking statins. (Lines 6 and 7)

(3)

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(c) (i) Using the results of the study, calculate how many times greater the score for symptoms when taking statins is compared with the score when not taking tablets.

(1)

times greater = ...

1. Question 4 · part a · 2 marks · Working and answer

Working and answer

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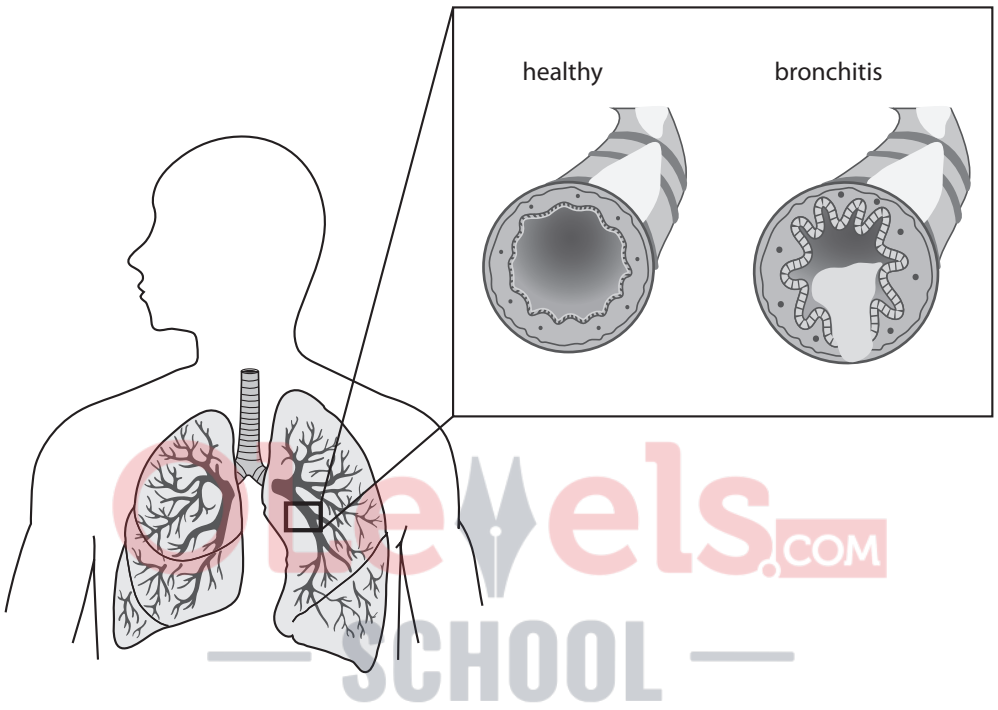
2. Question 6 · part c.iii · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.iii only.

6 Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.



(a) Name the part of the lungs affected by bronchitis.

(1)

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2. Question 6 · part c.iii · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.iii only.

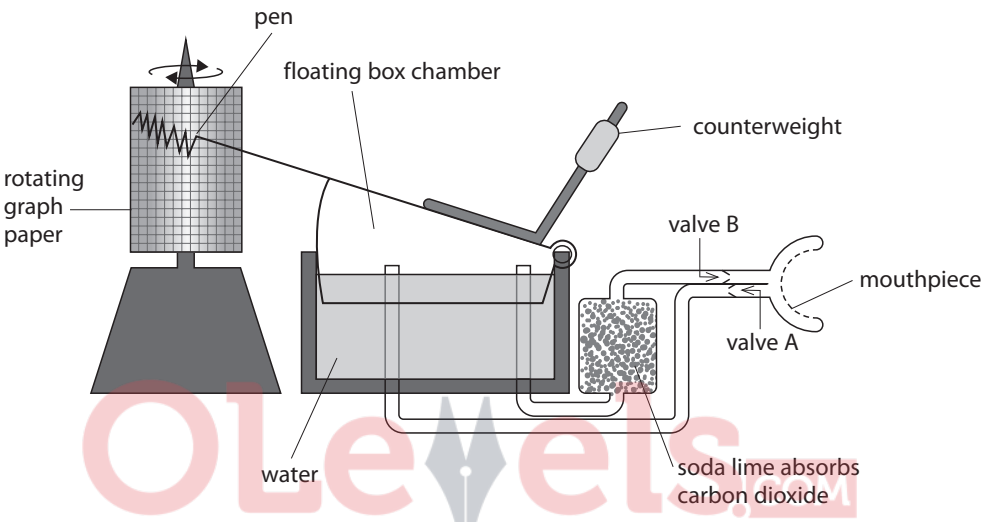
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(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.



(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

(4)

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2. Question 6 · part c.iii · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.iii only.

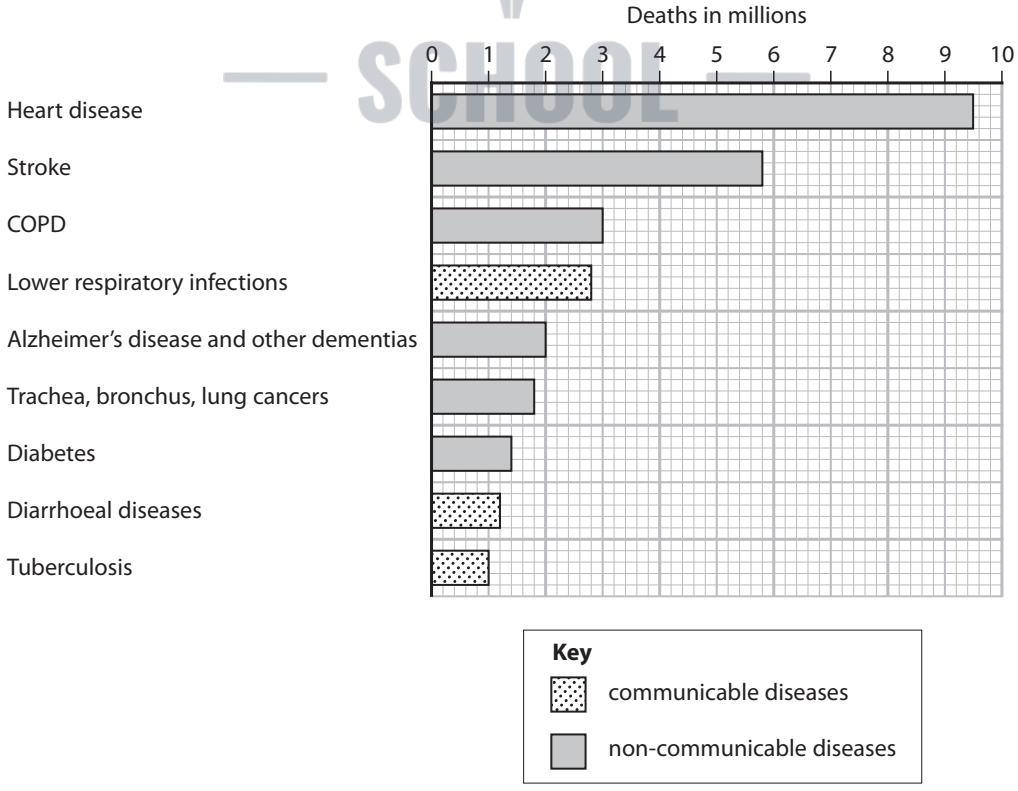
(ii) State one way to reduce risk to health when using a spirometer.

(1)

(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

(2)

(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.



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2. Question 6 · part c.iii · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.iii only.

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(i) More deaths are caused by non-communicable diseases than by communicable diseases.

Suggest what is meant by the term **non-communicable disease**.

(1)

.....

.....

.....

(ii) In 2000, the number of deaths from heart disease was 7.0 million.

Calculate the percentage change in the number of deaths from heart disease from 2000 to 2016.

(3)

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percentage change = ...%

(iii) Suggest a reason for the change in the number of deaths from heart disease from 2000 to 2016.

(1)

.....

.....

.....

(Total for Question 6 = 13 marks)

2. Question 6 · part c.iii · 1 mark · Working and answer

Working and answer

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3. Question 4 · part c.i · 2 marks

November 2024 Paper 2

Further practice · Attempt part c.i only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

malaria

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3. Question 4 · part c.i · 2 marks

November 2024 Paper 2

Further practice · Attempt part c.i only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people = ...

3. Question 4 · part c.i · 2 marks · Working and answer

Working and answer

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4. Question 4 · part c.ii · 4 marks

November 2023 Paper 2

Further practice · Attempt part c.ii only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Atorvastatin belongs to a group of medicines known as statins. Statins are medicines that regulate lipid levels in the blood. Statins are used when changes of diet and lifestyle have failed to lower blood cholesterol levels.

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(2)

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4. Question 4 · part c.ii · 4 marks

November 2023 Paper 2

Further practice · Attempt part c.ii only.

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(b) (i) Explain why people with liver disease or kidney problems should avoid taking statins. (Lines 5 and 6)

(4)

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(3)

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(c) (i) Using the results of the study, calculate how many times greater the score for symptoms when taking statins is compared with the score when not taking tablets.

(1)

times greater = ...

4. Question 4 · part c.ii · 4 marks

November 2023 Paper 2

Further practice · Attempt part c.ii only.

(ii) Discuss the results of the study using information from the table.

(4)

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(Total for Question 4 = 14 marks)

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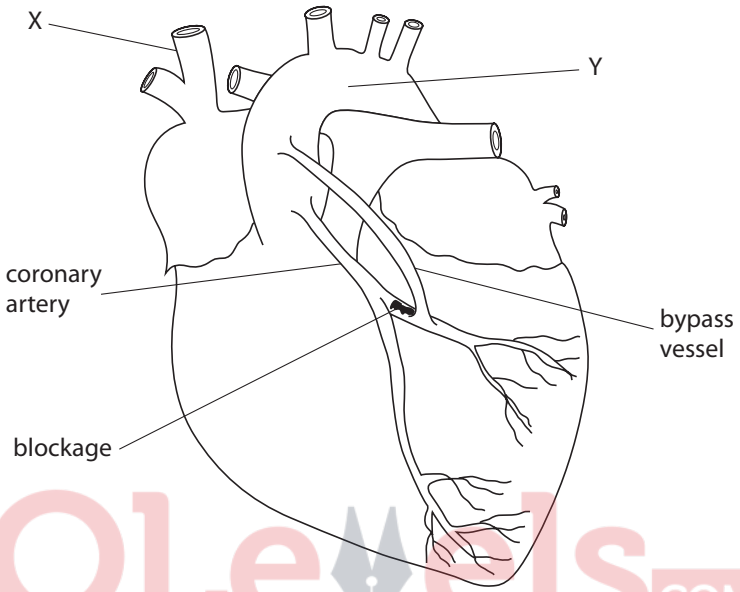
Working and answer

5. Question 3 · part b.iii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

3 The diagram shows a person's heart. The person has had a heart bypass operation because a coronary artery had become partly blocked.



(a) (i) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) To which part of the heart is blood vessel X connected?

(1)

- ☒ **A** left atrium
- ☒ **B** left ventricle
- ☒ **C** right atrium
- ☒ **D** right ventricle

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5. Question 3 · part b.iii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

(b) (i) Before the bypass operation, the coronary artery was becoming blocked at the point shown on the diagram.

Draw a circle on the diagram to show the area of the heart that was affected by the blockage before the bypass operation.

(1)

(ii) Describe what effect this blockage would have on the heart tissue if the bypass vessel was not inserted.

(3)

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(iii) Suggest how the bypass vessel allows the heart to work more effectively.

(2)

(Total for Question 3 = 8 marks)

5. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

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6. Question 9 · part b · 4 marks

June 2022 Paper 1R

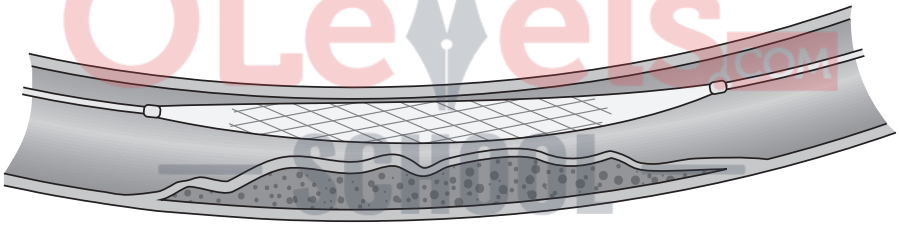
Further practice · Attempt part b only.

9 (a) Statins can be used to treat people with circulatory disorders.

Design a method that can be used to test the effectiveness of statins in treating circulatory disorders.

(3)

(b) The diagram shows a stent being used in the treatment of heart disease.



(Source: © DesignPrax/Shutterstock)

Explain how a stent can be used to treat heart disease.

(4)

(Total for Question 9 = 7 marks)

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6. Question 9 · part b · 4 marks · Working and answer

Working and answer

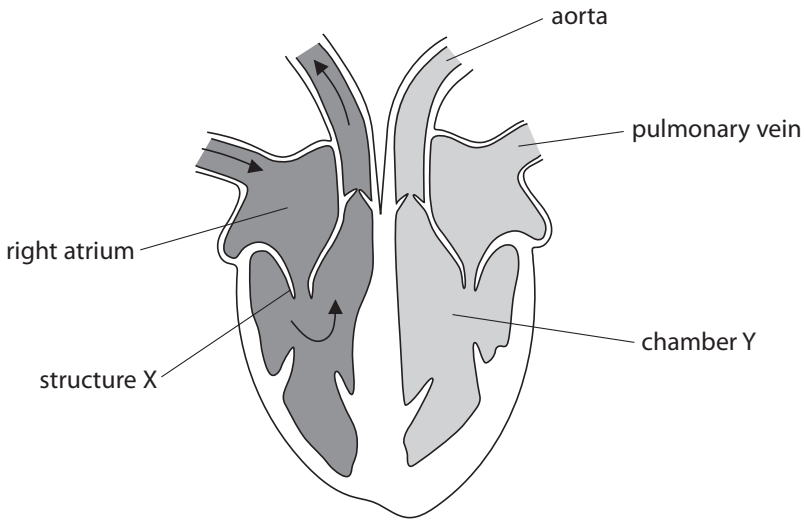
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7. Question 2 · part b · 2 marks

January 2023 Paper 2

Further practice · Attempt part b only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

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7. Question 2 · part b · 2 marks

January 2023 Paper 2

Further practice · Attempt part b only.

(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

(2)

Properties of blood	Blood in the aorta
oxygenated	
deoxygenated	
high pressure	
medium pressure	
low pressure	

(iv) State where blood in the left side of the heart has come from.

(1)

.....

(b) In one year in the United Kingdom, there were 173 heart transplants.

Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

(2)

1 .....

.....

2 .....

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7. Question 2 · part b · 2 marks

January 2023 Paper 2

Further practice · Attempt part b only.

(c) (i) Which blood component causes blood clotting?

(1)

- ☒ **A** lymphocytes
- ☒ **B** phagocytes
- ☒ **C** platelets
- ☒ **D** red blood cells

(ii) The diagram summarises part of the process that causes a blood clot.



Anticoagulant drugs prevent a blood clot forming.

These drugs stop the enzyme thrombin being produced.

Describe how a lack of thrombin prevents a blood clot forming.

(2)

(Total for Question 2 = 11 marks)

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7. Question 2 · part b · 2 marks · Working and answer

Working and answer

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